An Empirical Study of Adversarial Examples on Remote Sensing Image Scene Classification

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***Abstract*— Deep neural networks (DNNs), which learn a hierarchical representation of features, have shown remarkable performance in big data analytics of remote sensing. However, previous research indicates that DNNs are easily spoofed by adversarial examples, which are crafted images with artificial perturbations that fool DNN models toward wrong predictions. To comprehensively evaluate the impact of adversarial examples on the remote sensing image (RSI) scene classification, this study tests eight state-of-the-art classification DNNs on six RSI benchmarks. These data sets include both optical and synthetic-aperture radar (SAR) images of different spectral and spatial resolutions. In the experiment, we create 48 classification scenarios and use four cutting-edge attack algorithms to investigate the influence of the adversarial example on the classification of RSIs. The experimental result shows that the fooling rates of the attacks are all over 98% across the 48 scenarios. We also find that, for the optical data, the seriousness of the adversarial problem has a negative relationship with the richness of the feature information. Besides, adversarial examples generated from SAR images are used easily for fooling the models with an average fooling rate of 76.01%. By analyzing the class distribution of these adversarial examples, we find that the distribution of the misclassifications is not affected by the types of models and attack algorithms—adversarial examples of RSIs of the same class cluster on fixed several classes. The analysis of classes of adversarial examples not only helps us explore the relationships between data set classes but also provides insights for further designing defensive algorithms.**

***Index Terms*— Adversarial examples, deep neural networks (DNNs), remote sensing image (RSI) scene classification.**

I. Introduction

DURING the past several decades, remote sensing satellites have brought a tremendous amount of remote sensing images (RSIs) with various spectral, temporal, and spatial resolutions [1]. The accurate detection of ground objects, including buildings, roads, vehicles, trees, and others from RSIs, has been widely recognized as one of the most important tasks in remote sensing research [2].

In recent years, deep neural networks (DNNs) have been a standard technology in the remote sensing community because of their accuracy and efficiency in the feature extraction of RSIs [3]–[5]. DNNs can learn a hierarchical structure of features from RSIs via convolutional layers [6]–[8]. Shallow convolutional layers extract texture features of RSIs, while deep layers would learn the semantic features. A fully connected layer is used as a classifier to generate the prediction result. In the literature of RSI scene classification, DNNs have achieved sufficient progress, including novel model structure [9], [10], optimal algorithm [11], [12], and different data fusion strategies [13], [14]. DNNs have played an essential role in various tasks of RSI scene classification. Compared to traditional algorithms of RSI scene classification, DNNs are advantageous by providing end-to-end training models. It not only eliminates the problematic process of feature extraction but also achieves cutting-edge accuracies.

However, previous research also indicates several potential security problems existing in DNN models [15]–[18]. Szegedy [19] designed a particular small perturbation to attack a DNN model, and the DNN model will give a wrong prediction with a 99% confidence level, while humans are not even aware of it. These perturbed images are called adversarial examples. In Fig. 1, we show the workflow of adversarial examples generation. It can be divided into two stages. First, we need to train a model. It is the same as training the classification model. The weight of the model is updated through the gradient descent algorithm to reduce the loss. After the model converges, we add an adversarial perturbation to the input image and forward it into the model. The adversarial perturbation is updated through the gradient ascent algorithm to increase the loss. When the modified image turns the originally classified correctly image into another class, the modified image becomes an adversarial example. The adversarial example problem reduces the robustness of DNNs and undermines the capacity of feature extraction in DNNs. Therefore, the majority of RSI scene classification models using DNNs are under threat from adversarial examples that have drawn much attention from the remote sensing community [20], [21]. For example, in the military field, aircraft in the RSIs might be fooled as birds via attack algorithms [22]. Most of the studies of adversarial examples are based on natural images. Compared to natural images, RSIs

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have some unique characteristics. Most of them have a nadir perspective, without a clear foreground and background. They also have rich spectral and spatial information. It presents an urgent need for research in exploring the adversarial example problems of RSIs.

In this study, we extensively investigate the adversarial example problem in RSI scene classification. We select eight state-of-the-art DNN models that serve as the backbones of many detections and segmentation algorithms because of their superior performance in different remote sensing applications. These selected models, including AlexNet [23], VGG16 [24], ResNet50 [25], Inception-V4 [26], Inception-ResNet [26], ResNeXt [27], DenseNet-121 [28], and PNASNet [29], have various architectures, layer compositions, and classification performances over different remote sensing data sets. In our experiment, we applied these models to six different RSI data sets, including four optical and two synthetic-aperture radar (SAR) data sets with different spectral and spatial. We generate 48 classification scenarios based on the different models and data sets. Then, we apply four cutting-edge attack algorithms, including the fast gradient sign method (FGSM) [30], the basic iterative method (BIM) [31], DeepFool, and Carlini & Wagner (C&W) [32] to the 48 classification scenarios. Finally, 192 attack scenarios of experiments are conducted to analyze the adversarial example problem of RSI scene classification. The codes to reproduce our results are available on GitHub.[1]

From the experimental results, we find that the adversarial example problem is severe and widely existing in the tasks of RSIs classification. The fooling rates of the 48 classification models are all above 98%, demonstrating the vulnerability of the models on adversarial examples. To measure the change from regular samples to adversarial examples, we use L_2 perturbation distance to measure the magnitude or distance of change. For optical images, when the data set contains rich spectral bands, large sample sizes, high spatial dimensions, and spatial resolutions, attack algorithms would only need to modify very few pixels to fool the models. However, this propriety does not apply to SAR data sets. For SAR images, adversarial examples obtained from one model are more easily transferred to other models, which shows an average fooling rate of 76.01%. These conclusions reveal that SAR and optical images are quite different in adversarial example problems. From the observation of the class distribution of adversarial examples, we find that the misclassified classes are barely affected by the variety of models and attack algorithms. These adversarial examples from the same class have a similar distribution. The contributions of this article can be summarized as follows:

1) This article conducts a massive examination of the adversarial example problem in RSI scene classifications with richer RSI data sets, DNNs, and attack algorithms, which has not been fully recognized and resolved by previous research. We find that all the benchmark RSIs can be easily fooled at a rate of 98% or above.
2) We find that different types of optical RSIs reveal that the richness of feature information (i.e., image size, spatial resolution, and so on) is negatively correlated with adversarial perturbations, which indicates that RSIs with rich feature information are vulnerable to attack.
3) We find that adversarial examples generalize very well on SAR images among different DNN models. In six different RSI data sets, adversarial examples from SAR images are easier to be fooled and transferred than those generated on optical images with an average fooling rate of 76.01%.
4) We find that the types of models and attack algorithms have little effect on the classes of adversarial examples. The misclassified classes from the same class are highly clustered, which suggests that the class distribution of RSI data sets might be critical to design attack or defense algorithms accordingly.

In Section II, this article demonstrates the details for the data preparation and experimental design. Section III gives the results and analysis of our experimentation. Finally, conclusions and discussions are given in Section IV.

II. Preliminaries and Scenarios

A. Data Sets

In this study, we select six RSI data sets as our benchmarks for the following reasons.

1) *Diverse Data Types:* The benchmark data sets are collected from different areas using various sensors, such as optical, multispectral, and SAR images, with different resolutions and patch sizes.
2) *Various Classes of Ground Objects:* The data sets contain a variety of common ground objects, such as squares, airports, and parks. These ground objects are also the main objects in remote sensing classification applications. The selected RSI benchmarks are listed as follows.

The AID data set [33] is a large-scale aerial image data set from Google Earth imagery. It has several 10 000 images within 30 classes from the original optical aerial images, and all images are collocated from different remote imaging sensors.

The UC Merced Land Use Data set (UCM) [34] has 21 classes for various urban areas around the country. Each class has 100 images with 256×256, which are extracted from large images from the USGS National Map Urban Area Imagery collection.

The NWPU-RESISC45 (NWPU) data set [35] is created by Northwestern Polytechnical University and used for RSI scene classification. It covers 45 classes with 700 images in each class.

The EuroSAT-MS data set [36] has ten classes. The Sentinel-2 satellite system collects these images with 13 spectral bands ranging from 550 to 1342 nm. The dimension of the image is 64×64.

The MSTAR data set [37] is an SAR data set with a resolution of 0.3 m. This data set is used for the detection of vehicles, including 2S1 (self-propelled howitzer), BRDM2

[1] https://github.com/RayleighChen/AEs-remote-sensing-images

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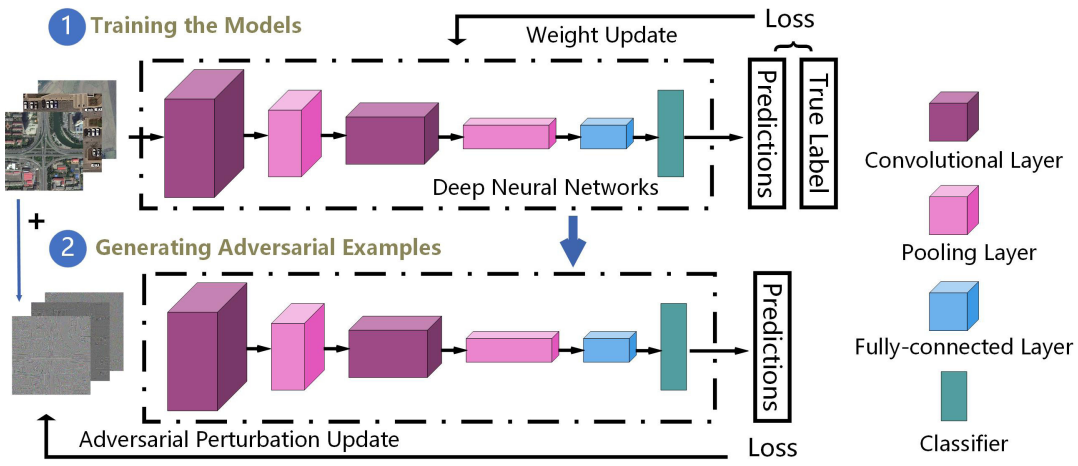


Fig. 1. Workflow of adversarial examples of RSIs. The whole process involves two stages. 1: training the model. It is the same as training a classification model. 2: generating adversarial examples. We input the RSIs with adversarial perturbation into the trained model from stage 1. According to the output loss, the adversarial perturbation is continuously updated through the gradient ascent algorithm. Modified RSIs drive the model to produce a greater loss. When the prediction is different from the true label, an adversarial example is generated.

TABLE I
METADATA OF THE DATA SETS

datasets	# classes	# per class	resolution(m)	image type	image size	year
AID	30	200∼400	0.5∼0.8	RGB	600×600×3	2017
UCM	21	100	0.3	RGB	256×256×3	2010
NWPU	45	700	∼30 to 0.2	RGB	256×256×3	2017
EuroSat-MS	10	3000,2500,2000	10	13 Bands	64×64×13	2017
MSTAR	10	450∼550	0.3	SAR	158×158	1995
SEN1-2 (SAR-Summer)	49	900∼1100	5	SAR	256×256	2018

(armored reconnaissance vehicle), T62 (tank), and D7 (bulldozer). In this study, we select ten classes from the data set for model evaluation.

The SEN1-2 data set [38] is a data set collected by the Sentinel-1 and Sentinel-2 for global observation in all seasons. In this study, we select the Sentinel-1 SAR images from the summer data set with 49 classes.

The metadata of these data sets is shown in Table I. In the experiments, we randomly choose 80% of the images for training and use the remaining 20% for validation.

TABLE II
DNN MODEL OVERVIEW

Models	# layers	# parameters	Year
AlexNet	8	60 million	2012
VGG16	16	138 million	2014
ResNet50	50	25.7 million	2016
Inception-V4	76	42.9 million	2016
Inception-ResNet	96	56.1 million	2016
ResNeXt	101	138 million	2017
DenseNet-121	121	8.2 million	2017
PNASNet	-	86 million	2018

B. DNN Models

DNNs consist of many hidden layers that learn a hierarchical representation of features. To be specific, these hidden layers contain three major structures: convolutional, pooling, and fully connected layers. The input data are initially fed to the convolutional layer, and then, the convolutional layer would extract features through different sizes of convolutional kernels. The outputs are called feature maps and subjected to nonlinear activation functions. The pooling layer would reduce the dimension of feature maps with the operation of max or mean sampling. A dense feature vector is produced by flattening the final feature maps by conducting the convolutional and pooling process several times. The dense vector is then fed into a fully connected layer to generate the prediction result. In the training process, backpropagation is adopted to update the parameters by minimizing the loss value calculated based on the predicted and reference labels. DNNs have shown their strong learning ability and proved to be successful in various domains of research, including pattern classification [39], computer vision [23], [29], and natural language processing [40]. DNNs have also been introduced to the remote sensing community for RSI analysis, such as pixel-based classification [2], object detection [41], and scene understanding [35].

For the remote sensing community, the basic architectures of DNNs are inspired by the breakthroughs in the development history of DNNs [10], [12], [14], including AlexNet, VGG, and Inception family. These models vary in different structures of layers. In this study, we select eight robust classification models, and the details are given as follows.

AlexNet [23] won the 2014 ImageNet competition and is the most well-known DNN. It is one of the pioneers of DNN models. AlexNet is composed of eight layers, including five pairs of convolutional layers followed by two fully connected

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layers and a classification layer. Starting from AlexNet, DNNs begin to draw the attention of academia in various applications.

VGG16 [24] is derived from AlexNet. It contains a deeper structure of layers (16) than AlexNet. Instead of using a big kernel size like in AlexNet, VGG16 uses 3×3 kernels and proves to be effective. Previous research shows that VGG networks can enhance the effective learning of a hierarchical representation of features and improve classification accuracies.

ResNet50 [25] belongs to the ResNet family. ResNet uses a residual block to solve the problem of vanishing gradients. The residual block helps to connect shallow data to deeper layers via an identity shortcut connected. The performance of ResNet outperforms human classification accuracy on the ImageNet data set.

Inception-V4 [26] comes from the family of inception networks. The first network of inception family is GoogleNet (Inception-V1) [42]. It introduces a new module named inception to extract multilevel features by changing convolutional kernel sizes. The multilevel features generated are stacked along the channel dimension as the input to the next layer. By adding inception module, many popular models from this family emerge, including Inception-V1 [42], Inception-V2 [43], Inception-V3 [43], and Inception-V4. In general, the models from the inception family have deeper layers but with significantly fewer weights. In this study, we select Inception-V4 as the representative of the inception family.

Inception-ResNet [26] is also a model from the inception family. It combines inception and ResNet to speed up the training process by using the residual module.

ResNeXt [27] is an improvement of Inception through the cardinality. The cardinality is referred to as the size of the set of transformations. For the inception family, layers from the inception module need to be customized separately due to a diversity of kernel sizes. ResNeXt utilizes the inception module to split, transform, and merge blocks. Through multiple transformations, the model can improve accuracy without adding the learning weights and reduce the number of hyperparameters.

DenseNet-121 [28] is also proposed to solve the problem of vanishing gradient. It expands the identity shortcut connected to connect all layers within a dense block. This structure improves the flow of information throughout all layers, which relieves overfitting and the learning weights. In addition, DenseNet adopts the cross-layer connectivity to enable feature reuse in the inference process.

PNASNet [29] is a product of Network architecture Search (NAS) technology [44], which is used for generating DNNs automatically. PNASNet achieves state-of-the-art performance in the ImageNet challenge.

The summary of all models is shown in Table II. We separately train eight DNN models using the six RSI data sets. During model training, we use the fine-tuning strategy from the pretrained models of the ImageNet data sets.

C. Attack Algorithms

The small adversarial perturbations coming from attack algorithms can confuse the trained DNNs to produce incorrect predictions over those initially corrected predicted ones. Furthermore, adversarial examples can fool multiple DNNs. Adversarial examples showed that the DNNs are susceptible to attack algorithms, which might be imperceptible to human vision [15], [45].

We are given a trained DNN model f and original unmodified input RSI x with the corresponding label y. We define the adversarial example and misclassified class as x' and y'. The misclassified class is the predicted label of an adversarial example. ρ represents the adversarial perturbation. The adversarial example problem of RSI scene classification can be defined as a box-constrained optimization problem as follows:

$$\begin{aligned} &\min_{\rho} \|\rho\| \\ &\text{s.t. } \rho = \|x' - x\|, \\ &\quad f(x) = y, \\ &\quad f(x') = y', \\ &\quad y \neq y' \end{aligned} \tag{1}$$

where $\|\cdot\|$ denotes the distance between two inputs. The optimization problem minimizes the adversarial perturbation ρ, where the RSI x added ρ can fool the prediction of DNNs and is almost imperceptible to human eyes. In this study, we select four cutting-edge attack algorithms to generate adversarial examples x'.

For generating adversarial examples, we use the FoolBox tool [46] to fool the trained DNNs. We use an Ubuntu 16.04 machine with a 12×3.20 GHz Core i7 CPU and NVIDIA GTX 2080Ti GPU for the experimentation.

1) Fast Gradient Sign Method: To develop an effective attack algorithm, Goodfellow proposed a method called FGSM [30] to compute the adversarial perturbation instead of using the expensive linear search method. This method is designed based on the idea of stochastic gradient descent and is shown as follows:

$$x' = x + \epsilon \text{sign}(\nabla J(\theta, x, y)) \tag{2}$$

where $\nabla J(\cdot, \cdot, \cdot)$ represents the gradient of the loss function, and θ represents the weights of DNNs. $\text{sign}(\cdot)$ denotes the sign function, and ϵ denotes the small magnitude of the perturbation. The adversarial perturbation $\rho = \epsilon \text{sign}(\nabla J(\theta, x, y))$ is restricted by the normal scalar ϵ.

The perturbation changes the image along the direction of the gradient at each pixel, increasing the loss value of DNNs to get an incorrect result. FGSM is also referred to as one step attack algorithm since it conducts gradient update one step at a time.

2) Basic Iterative Method: In contrast to taking the one-step update in the direction of the gradient in loss, an extension that performs multiple small steps in the gradient descent is proposed by Kurakin, which is named the BIM [31]. This method conducts smaller changes in multiple iterations as follows:

$$x'_{i+1} = \text{Clip}\{x'_i + \epsilon \text{sign}(\nabla_X J(\theta, x, y)\} \tag{3}$$

where i represents the number of iterations, and Clip can remain adversarial examples in the range of value. The initialization adversarial example x'_i is the original input RSI x.

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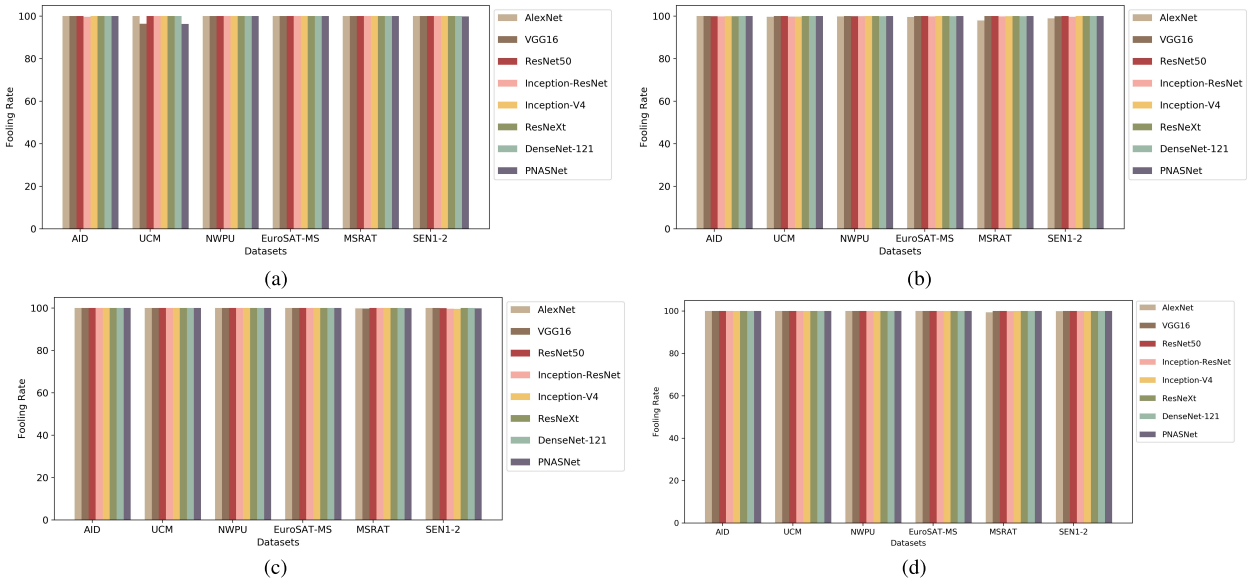


Fig. 2. Fooling rates in all attack scenarios. (a) FGSM. (b) BIM. (c) DeepFool. (d) C&W.

TABLE III
CLASSIFICATION ACCURACY OF THE DNNS

Models	AID	UCM	NWPU	EuroSAT-MS	MSRAT	SEN1-2
AlexNet	84.50%	81.42%	80.54%	93.18%	78.11%	63.61%
VGG16	93.75%	95.95%	91.81%	97.24%	85.61%	86.74%
ResNet50	92.50%	95.95%	91.73%	98.09%	79.34%	78.26%
Inception-V4	91.20%	92.38%	91.93%	97.79%	98.35%	97.59%
Inception-ResNet	89.70%	89.28%	91.00%	97.37%	98.59%	97.04%
ResNeXt	96.10%	97.85%	95.55%	98.64%	97.44%	93.71%
DenseNet-121	93.05%	94.28%	92.31%	98.13%	80.46%	70.63%
PNASNet	96.15%	97.61%	96.00%	98.14%	97.89%	99.79%

After several iterations, we can achieve the final adversarial example. Compared to FGSM, BIM is more effective in fooling DNNs, but it is less efficient in time for computing adversarial perturbations [31].

3) DeepFool: Moosavi-Dezfooli proposed DeepFool to compute adversarial perturbation in an iterative manner [47]. In general, we assume that DNNs can learn many decision boundaries to divide different classes in the data set. The DeepFool algorithm can make the input RSI cross the closest decision boundary to fool the models. The new data, an adversarial example, can be another incorrect class. For some DNNs with high nonlinearity, DeepFool adopts a linear approximation strategy to perform the iterative attack.

For a binary classifier $g(\cdot)$, it could linearize the decision boundary, such as $\mathcal{F} = \{x : w^T x + b = 0\}$. w and b are the weights of the classifier. The adversarial perturbation to the classifier can be denoted as $\rho^*(x) = -\frac{f(x)}{\|w\|2} w$. During each iteration, an iterative method is conducted to approximate the adversarial perturbation. In the final iteration, the model accumulates approximation from previous iterations to form a vector to the classification boundary. By moving along the vector, the adversarial perturbation can make the input image cross the decision boundary. The adversarial perturbation is defined as follows:

$$\begin{aligned} &\arg\min_{\rho_i} \|\rho_i\| \\ &\text{s.t. } g(x_i) + \nabla g(x_i)^T \rho_i = 0. \end{aligned} \tag{4}$$

This method can be easily extended to multiclass classifiers. In these DNNs, DeepFool performs fewer perturbations for adversarial examples.

4) Carlini and Wagner: The Carlini and Wagner (C&W) attack algorithm is proposed by Carlini and Wagner [32] following optimization problem as follows:

$$\min_{\rho} \|\rho\|_p + c \cdot p(x') \tag{5}$$

where p is the new function, $p(x') \geq 0$ if and only if $f(x') = y'$, and c is a hyperparameter used to trade off the relationship between two losses. The author also created an effective function for attacking, and it is shown as follows:

$$p(x') = \max(\max_{i \neq y'}(Z(x')_i) - Z(x')_t, -k) \tag{6}$$

where $Z(\cdot)$ denotes the Softmax function, and t is the class after the attack. k is the confidence of the output, which is a customized parameter. A high value of k indicates high confidence in the model, while it comes with adversarial examples. On the other hand, the C&W attack algorithm uses a w that satisfies $\rho = \frac{1}{2}(tanh(w) + 1) - x$ to transform

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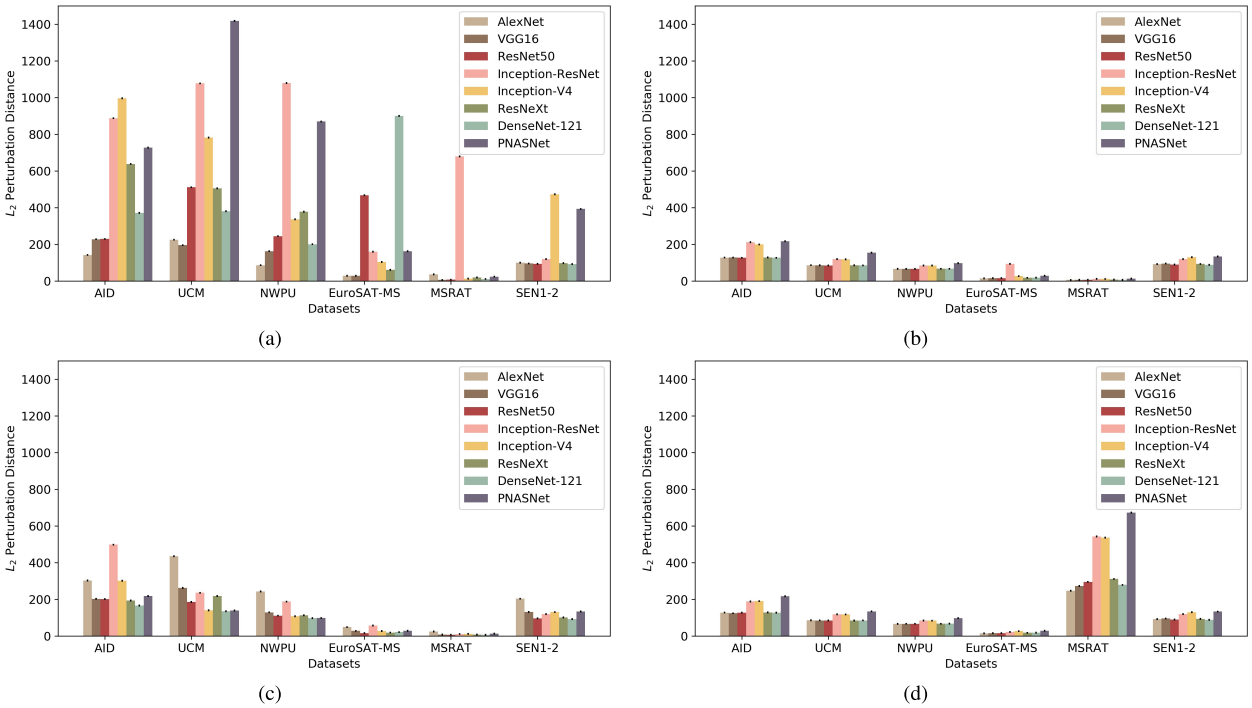


Fig. 3. L_2 perturbation distance in all attack scenarios. (a) FGSM. (b) BIM. (c) DeepFool. (d) C&W.

adversarial perturbation instead of constraining ρ shown in 1.

The adversarial perturbation generated from C&W is usually small, which can be used to attack many defensive models [32]. However, the efficiency of the C&W attack algorithm is very low.

D. Evaluation Metric

We use two evaluation metrics to analyze the effect of attack algorithms: the fooling rate and the L_2 perturbation distance.

1) Fooling Rate: The fooling rate is a ratio of adversarial examples that would fool the DNN model in all attack images. It can be represented as follows:

$$r = \frac{n_{\text{fooled images}}}{n_{\text{attack images}}} \tag{7}$$

where $n_{\text{fooled images}}$ represents the incorrect labels generated from adversarial examples, and $n_{\text{attack images}}$ represents all the attack images. A higher fooling rate indicates the higher vulnerability of the DNN model or a stronger attack ability of adversarial examples. Thus, the fooling rate would be an important indicator to evaluate the robustness of the DNNs.

2) Perturbation Distance: We use perturbation distance to measure the difference between the original input and the adversarial example. Perturbation distance can be used to evaluate the cost of generating adversarial examples [32], [48], [49]. In this study, we apply L_2 norm of the two images to compute the perturbation distance. It can be written $\|x' - x\|_2$, where the L_2 norm $\|\cdot\|_2$ is defined as

$$\|v\|_2 = \left(\sum_{i=1}^{n} |v_i|^2\right)^{\frac{1}{2}}. \tag{8}$$

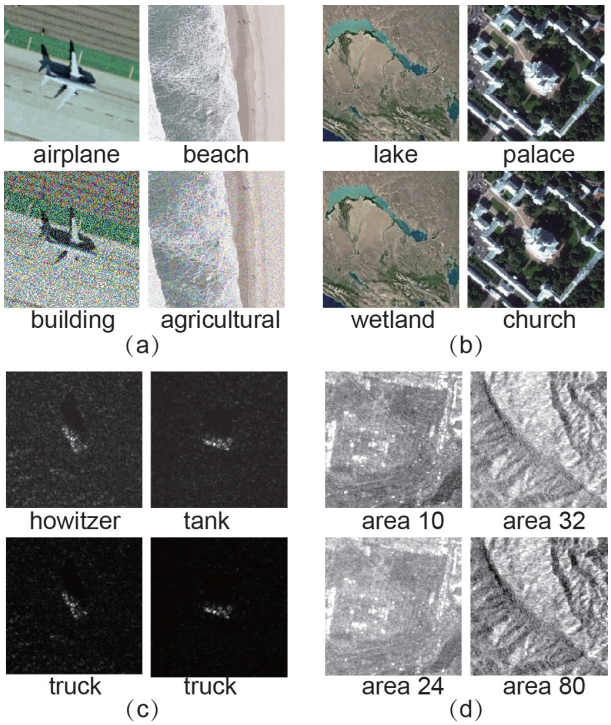


Fig. 4. Original images and adversarial examples are paired up and down. The upper images are original ones, and the lower images are adversarial examples. (a) Comes from PNASNet with FGSM using the UCM data set. (b) Comes from DenseNet-121 with C&W using the NWPU data set. (c) Comes from AlexNet with Deepfool using the MSRAT data set. (d) Comes from Inception-V4 with BIM using the MSRAT data set.

A large distance indicates that an attack algorithm needs to modify more pixels to fool a DNN model, which means that an attack algorithm is weak or a DNN model is robust.

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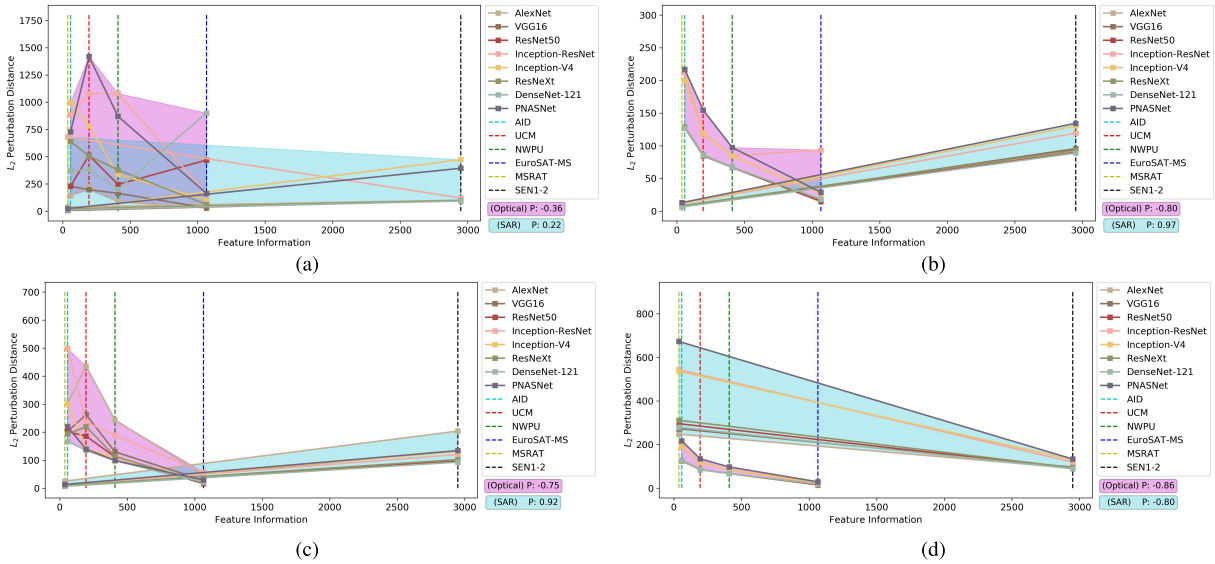


Fig. 5. Correlation between the feature information and L_2 perturbation distance. We use different colors to show the trend of optical and SAR data sets under different attack algorithms and DNN models. The trend refers to the correlation between adversarial perturbation and feature information. (a) FGSM. (b) BIM. (c) DeepFool. (d) C&W.

III. Experiments

A. Classification Benchmark

We build remote sensing image classification scenarios-48 using eight representative DNNs models and six RSI data sets. For the training model, the learning rate of all models is 0.0001, and their batch size is 32. All models have been trained for 50 epochs. The classification accuracies are shown in Table III.

From Table III, we can see that the DNNs perform differently on different types of RSIs. The average accuracy of all models is 91.55%. In 32 classifications of the optical data sets, 27 of the models generate an accuracy of 90% or above, and the rest scenarios have an accuracy of 80% or above. For optical data sets, the average accuracy of the models is 93.47%. For SAR data sets, eight models produce an accuracy of more than 90%, and the average accuracy is 87.69%. In general, the overall accuracy of DNNs on optical data sets is better than on SAR images.

AlexNet, which is recognized as the pioneer of DNNs, generates an average accuracy of 80% or above on all of the RSI data sets. However, AlexNet produces a worse result on the two SAR data sets, with an accuracy of 78.11% and 63.61%, respectively. This demonstrates the different properties of the optical and SAR data sets. For AlexNet, VGG16, ResNet50, and DenseNet-121, the accuracy of the optical data set is higher than that of SAR images. However, two Inception family models, including Inception-v4 and Inception-ResNet, perform especially good (97.89% on average) on the SAR data set. ResNeXt and PNASNet perform well on both optical and SAR data sets. Especially, PNASnet, which is automatically generated from the ImageNet data set, achieves the best accuracy (>96%) on all of the RSI data sets. The same model has different performance on different types of data sets. In the same data set, the latest model may not get the best performance. There is no uniform criterion for model classification performance. The relationship between model architecture and data is still a black box, which needs to be further studied.

Our benchmark includes the most advanced DNNs and multiple types of RSI data sets. The classification results generated from the DNN models are also helpful for analyzing the interactions between the models and RSIs; 48 trained DNN models contain diverse classification scenarios and are used to generate adversarial examples.

TABLE IV
Parameters of Attack Algorithms

Attack	ϵ	Learning Rate	Iterations
FGSM	0.3	-	-
BIM	0.3	-	10
DeepFool	-	-	100
C&W	-	0.005	100

B. Are Adversarial Examples Universal in RSI Scene Classification

In this section, we test the commonly existing adversarial example problem in RSI scene classification using four attack algorithms (FGSM, BIM, DeepFool, and C&W) to the RSICS-48. The fooling rate is used to measure the probability of success for an attack algorithm to generate adversarial examples. Adversarial examples are sampled from the validation set that the model predicts correctly. The hyperparameters about the attack algorithm are shown in Table IV. We compare the fooling rates of the attack algorithms by creating different classification scenarios, and the results are shown in Fig. 2.

Fig. 2 shows that the fooling rates from 192 attack scenarios (AS-192) are above 98%, where a majority of them produce a percentage of 100%. The experimental result reveals that the DNNs can be easily fooled by the attack algorithms regardless

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of the type, resolution, and size of the RSI data sets. The lower bound of AS-192 is 98%, which is generated by the FGSM attack algorithm on VGG16 and PNASnet models using the UCM data set. The experiment reveals that all the RSI classes are not safe, and the adversarial example is a severe and universal problem of RSI scene classification.

C. Characteristics of Adversarial Examples of RSIs

To evaluate the effect of adversarial examples, we utilize the L_2 perturbation distance to measure the distance between the original data and its adversarial examples in all of the AS-192. A higher L_2 perturbation distance indicates either a weaker attack algorithm or a more robust DNN model. The details are shown in Fig. 3.

Although all the attack algorithms generate a high fooling rate, the attack algorithms are different for the changed pixels of the original data according to the distribution of L_2 perturbation distance. For FGSM, adversarial examples generated from Inception-ResNet using the AID, UCM, and NWPU data sets need to alter more pixels than those using the EuroSAR-MS data set. This phenomenon also happens to the DenseNet-121 model. FGSM is the weakest attack algorithm among all the attack algorithms. Adversarial examples generated from FGSM always have a large L_2 perturbation distance from the original data. BIM and C&W have a similar performance by altering a limited number of pixels to fool the DNNs in most of the scenarios. However, they have dramatically different performances on the MSRAT data set. From the perspective of the SAR data set, the attack from C&W needs to pay a higher cost of attacking DNNs trained on MSRAT compared to other data sets, which indicates that it is more difficult for the C&W attack algorithm to generate adversarial examples. However, this difference might not come from the difference of sensors since they have nearly a consistent performance on the SEN1-2 data set. For the same data set and model, the impact of the attack algorithm shows a huge difference. The performances of different attack algorithms are quite diverse. The result indicates adversarial examples generated from the different attack algorithms have a high fooling rate (100%) and are quite different regardless of the types of models or data sets.

This difference in perturbation distance also exists in adversarial examples. Several adversarial examples are shown in Fig. 4. The two pairs of images on the upper left are from PNASNet with FGSM using the UCM data set. This attack scenario has a large L_2 perturbation distance shown on Fig. 3. The changed pixels are the noise presenting in adversarial examples generated by the attack algorithms to attack the DNN models successfully. The two pairs of images in the upper right are from DenseNet-121 with C&W trained using the NWPU data set. We find no obvious change of pixels on adversarial examples. The lower pairs are SAR images from the two SAR data sets generated from different attack algorithms and DNNs. The perturbations to the SAR images are normal and are hard to interpret for human eyes.

To further explore the relationship between the different types of data sets and attack algorithms, we compare the feature information and L_2 perturbation distance. The feature information of an RSI data set indicates the abundance of spectral and spatial, including the number of classes, the image size, and the spatial and spectral resolutions. In this study, we consider two main factors for defining feature information. The first factor is the feedforward process of the DNNs. When the image is input to the model, for the model, the dimension of the input image comes from the resolution and the number of channels of the image. Every pixel is a dimension. An image with a larger dimension has more space can be attacked. The second factor is the definition of adversarial examples. The number of classes provides more options for adversarial examples. Therefore, we multiply the value of these properties to get the feature information of the RSI data sets. The result is shown in Fig. 5.

To explore the relationship between the feature information and L_2 perturbation distance, we compute the Pearson correlation coefficient [50] of optical and SAR data sets, respectively. For the optical data sets, the Pearson coefficients are -0.36, -0.80, -0.75, and -0.86, respectively, among the four attack algorithms. Therefore, the L_2 perturbation distance is negatively correlated with the richness of feature information in the optical data sets. This characteristic also relates to attack algorithms. For FGSM, the correlation between feature information and L_2 perturbation distance is not reliable. From Figs. 3 and 4, we see that FGSM is the weakest attack algorithm, and it produces more adversarial perturbation. Therefore, FGSM always needs to change more pixels in most levels of feature information. In contrast, we know that the C&W attack is the strongest attack using the same level of feature information for RSI scene classification.

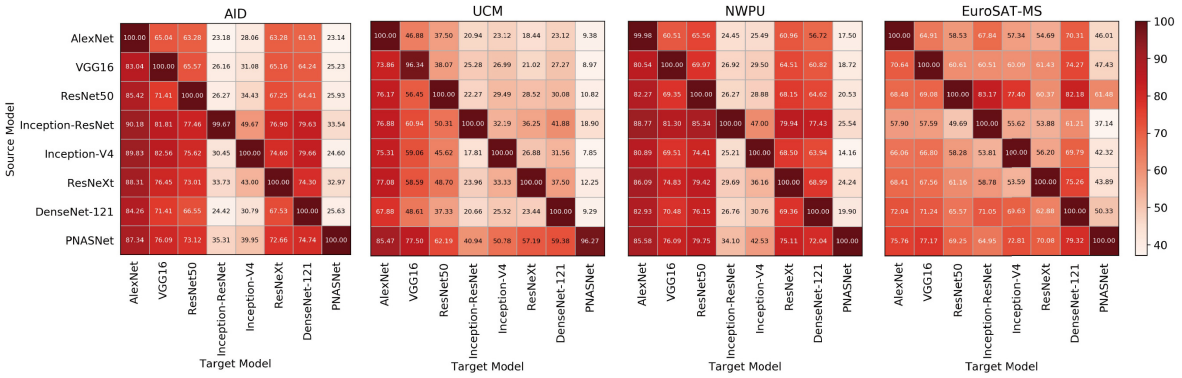
However, for SAR data sets, the Pearson coefficients are 0.22, 0.97, 0.92, and -0.80. Different from the consistent correlation of optical images, SAR data sets show an inconsistent correlation with the richness of feature information and adversarial perturbation. It may also be that the definition of feature information is not rigorous. We also look forward to finding better measurement methods in the future.

D. Transferability Benchmark

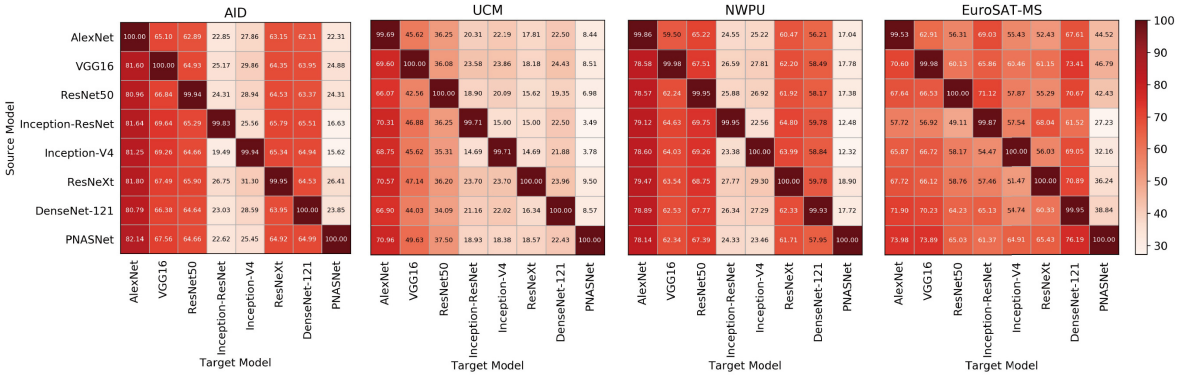
Adversarial examples have high transferability and can fool many different DNNs [48]. In this study, we use adversarial examples generated from a model and test them on other DNNs using the same RSI data sets. Figs. 6 and 7 show the transferability of adversarial examples in the different scenarios. The heatmap matrix shows the adversarial examples generated from the ith row model when applied to the jth column model. Values on the diagonal are the fooling rates of the original model. Different models have different requirements for the input size of data, which helps reduce the fooling rate of adversarial examples [51]. We resize the larger RSIs by cropping from the center while adding paddings to the smaller RSIs to fit the data in different models.

Adversarial examples have an excellent transferability to fool other models by providing that the fooling rates are above 60% in 68.49% of the cases. According to Figs. 6 and 7, attack algorithms have similar heatmap under the same type of RSI data sets, indicating that differences of attack algorithms have

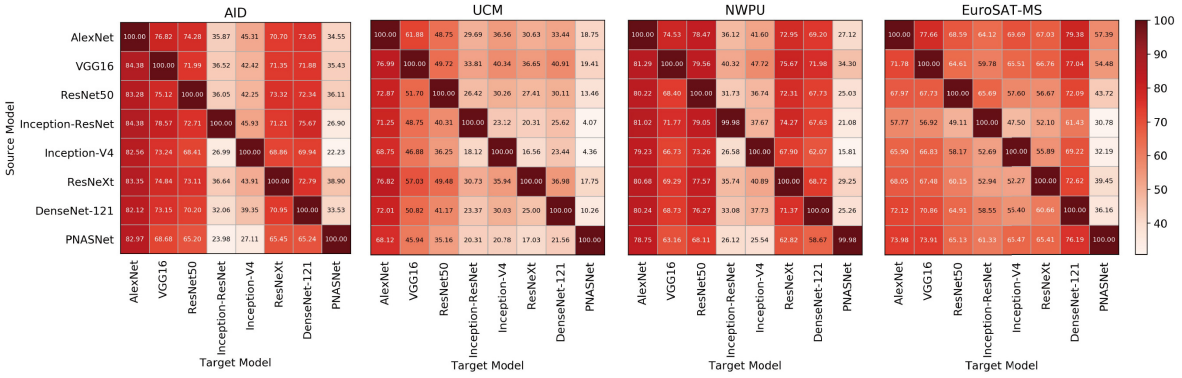
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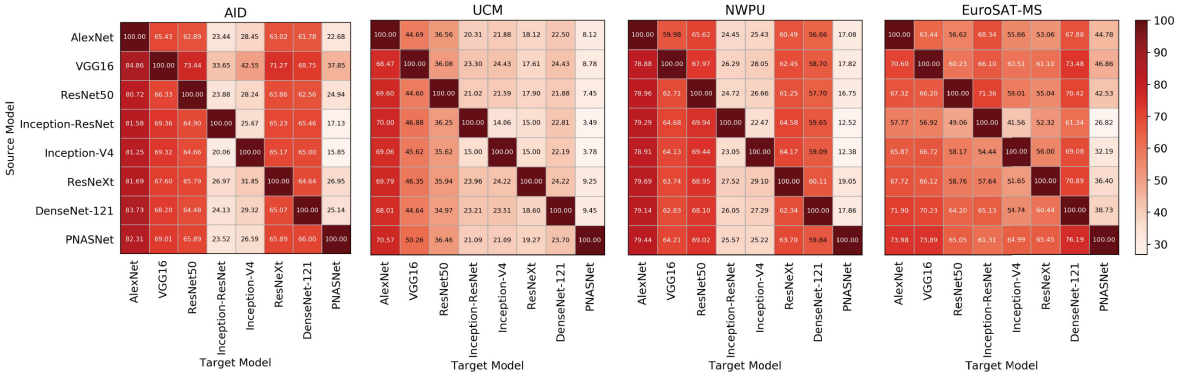
(a)



(b)



(c)



(d)

Fig. 6. Transferability of adversarial examples on optical data sets. We use adversarial examples obtained from the source models to attack the target models. (a) FGSM. (b) BIM. (c) DeepFool. (d) C&W.

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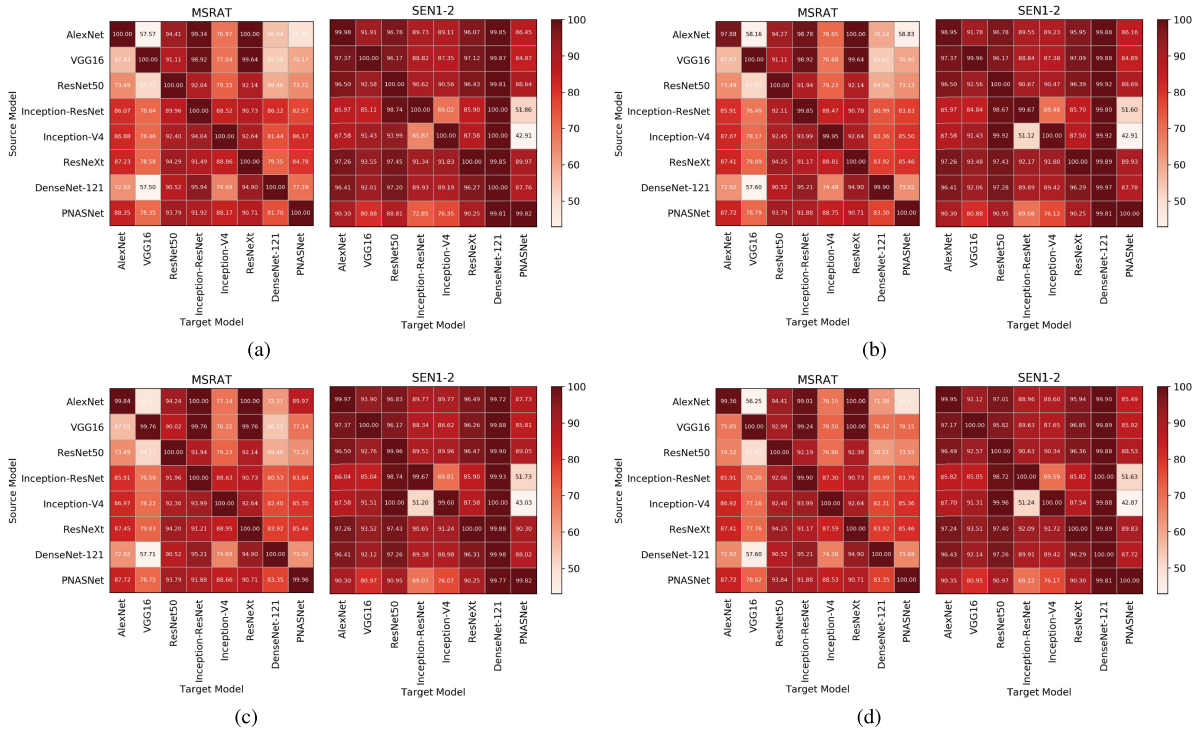


Fig. 7. Transferability of adversarial examples on the SAR data set. We use adversarial examples obtained from the source models to attack the target models. (a) FGSM. (b) BIM. (c) DeepFool. (d) C&W.

less impact on transferability, while the variety of RSI data sets is more critical for transferability. Adversarial examples generated from SAR data sets are more accessible to fool other models by making an average fooling rate of 76.01% compared to those produced on optical data sets, which has an average fooling rate of 60.28%. It shows that adversarial examples obtained on the SAR data set are more severe than the optical data sets.

PNASnet and inception family models, such as Inception-ResNet and Inception-V4, achieve significantly better defensive results than other models on the optical data sets. According to Fig. 3, attack algorithms usually need to change many pixels to fool these models. In contrast, AlexNet fails to classify almost all adversarial examples with the worst performance compared to other DNN models. For SAR data sets, DenseNet-121 is not suitable for classification tasks on SAR data sets since it is vulnerable to attack algorithms. The fooling rates of adversarial examples generated from DenseNet-121 are close to 100%. The PNASnet and inception family have the best defense ability across all of the data sets.

The heatmap matrix also reveals that the transferability of adversarial examples is asymmetric. For example, adversarial examples generated using the AID data set from the Inception-V4 via FGSM can fool AlexNet with a fooling rate of 81.25%. In contrast, adversarial examples from AlexNet can only fool Inception-V4 with the fooling rate of 27.86% under the same condition, which reveals that the transferability of adversarial examples is heavily influenced by the model used.

E. Class Distributions of Adversarial Examples

The class distribution of adversarial examples is diverse among the different data sets. To explore the relationship between RSI data sets and adversarial examples, we calculate the classes of adversarial examples generated by different attack algorithms and models with the same data set. We find that the misclassified classes are highly concentrated, and the distribution of adversarial examples is consistent within the same data set. As an example, we use a pie chart (as shown in Fig. 8) to visualize the distribution of classes of adversarial examples generated from the UCM data set.

Adversarial examples have high similarity in those misclassified classes despite different models and attack algorithms. Fig. 8(a) demonstrates the distribution of adversarial examples from the same DNN model using different attack algorithms. Each slice of the pie chart indicates the proportion of different types of adversarial examples. Different colors represent different ground object classes. The input RSIs are from the class of intersection, where the diversity of adversarial examples is large, including the classes of the overpass, building, and mobile airplane. Adversarial examples generated from Inception-ResNet occupy an average proportion of 82.68% among different attack algorithms in the classes of the overpass, building, airplane, and storage banks. Fig. 8(a) shows the misclassified distribution of Inception-ResNet, and the result is similar to DenseNet-121 models. The two models have nearly the same distribution of adversarial examples within a limited number of classes. On the other hand, the results from the different models using the same attack algorithms are shown in Fig. 8(b). These RSIs are also from the intersected class of the UCM data set. From Fig. 8(b), we can see that the overpass, storage banks, and building are the main classes of adversarial examples for the same attack algorithm with different models.

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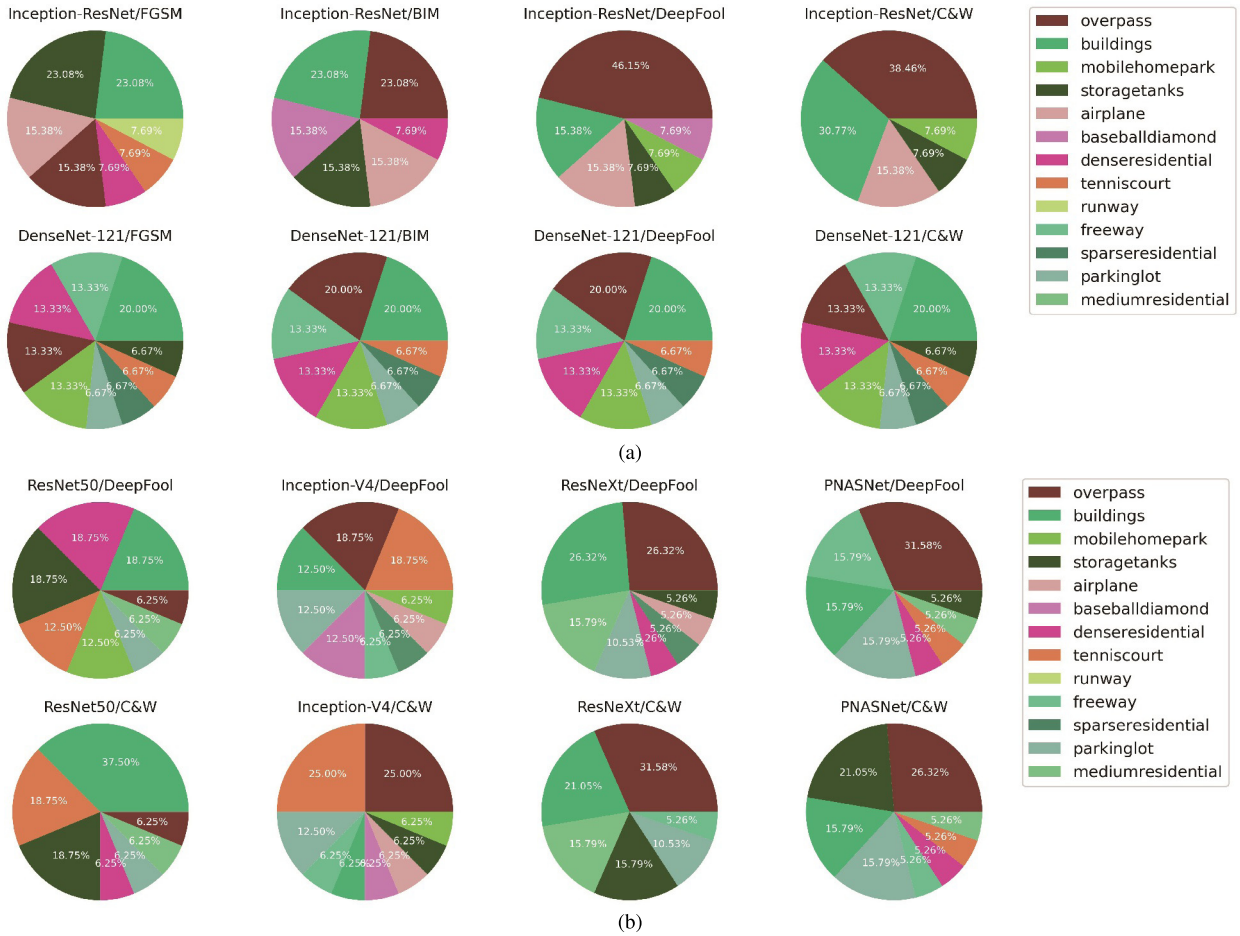


Fig. 8. Class distributions of adversarial examples in the class of intersection of the UCM data set. (a) Class distribution of the Inception-ResNet and DenseNet-121 in different attacks. (b) Class distribution of the DeepFool and C&W attack algorithm across different DNN models.

The properties of the class distribution of adversarial examples also are observed for SAR images. Fig. 9 shows the distribution of classes of adversarial examples from the SAR data set (MSTAR). All adversarial examples come from the class of D7 (bulldozer). Fig. 9(a) shows the class distribution from Inception-ResNet and DenseNet-121 in four attack algorithms, and Fig. 9(b) shows the class distribution of different DNNs with the same attack algorithm. We can find that whether adversarial examples come from different attack algorithms or different models, their misclassified classes are generally consistent. The ZIL131 (cargo truck) and the ZSU_23_4 (antiaircraft vehicle) are the main classes of adversarial examples for the original bulldozer class.

Above all, the experimental result shows that the classes of adversarial examples would barely be affected by the architecture of DNNs or attack algorithms. This phenomenon may be due to many factors: 1) the quantity and quality of images in the training data set, for example, if the data set has only two classes, adversarial examples can only be the other class, regardless of whether the two classes have a strong correlation and 2) homogeneity and heterogeneity between classes. An image has a greater chance to become an adversarial example of its similar classes. Chen *et al.* [52] found that the classes closer in the feature space are more likely to be the classes of adversarial examples. These factors need to be further explored. Meanwhile, we find that the class distribution of adversarial examples represents the relationship between the different classes of the data set. For example, the cargo truck is the main class of adversarial examples from the bulldozer class for the MSTAR data set. Thus, we can safely infer whether the cargo truck and bulldozer have a strong correlation or similarity in the feature space or not. This provides a novel way for us to explore the relationship among different data sets, as well as insights for designing defensive algorithms to protect the model.

F. Discussion

In the field of remote sensing, whether similar scenes would become another problem for adversarial examples, such as the palace and church, and their labels are difficult to distinguish from the perspective of satellites. Without adding any noise, the model may classify the RSI to another similar label. In fact, they are different from adversarial examples. When enough adversarial examples are generated, according to the experimental results in Section III-E, we find that the distribution of all adversarial example classes of a certain class is consistent in a variety of attack scenarios. For humans,

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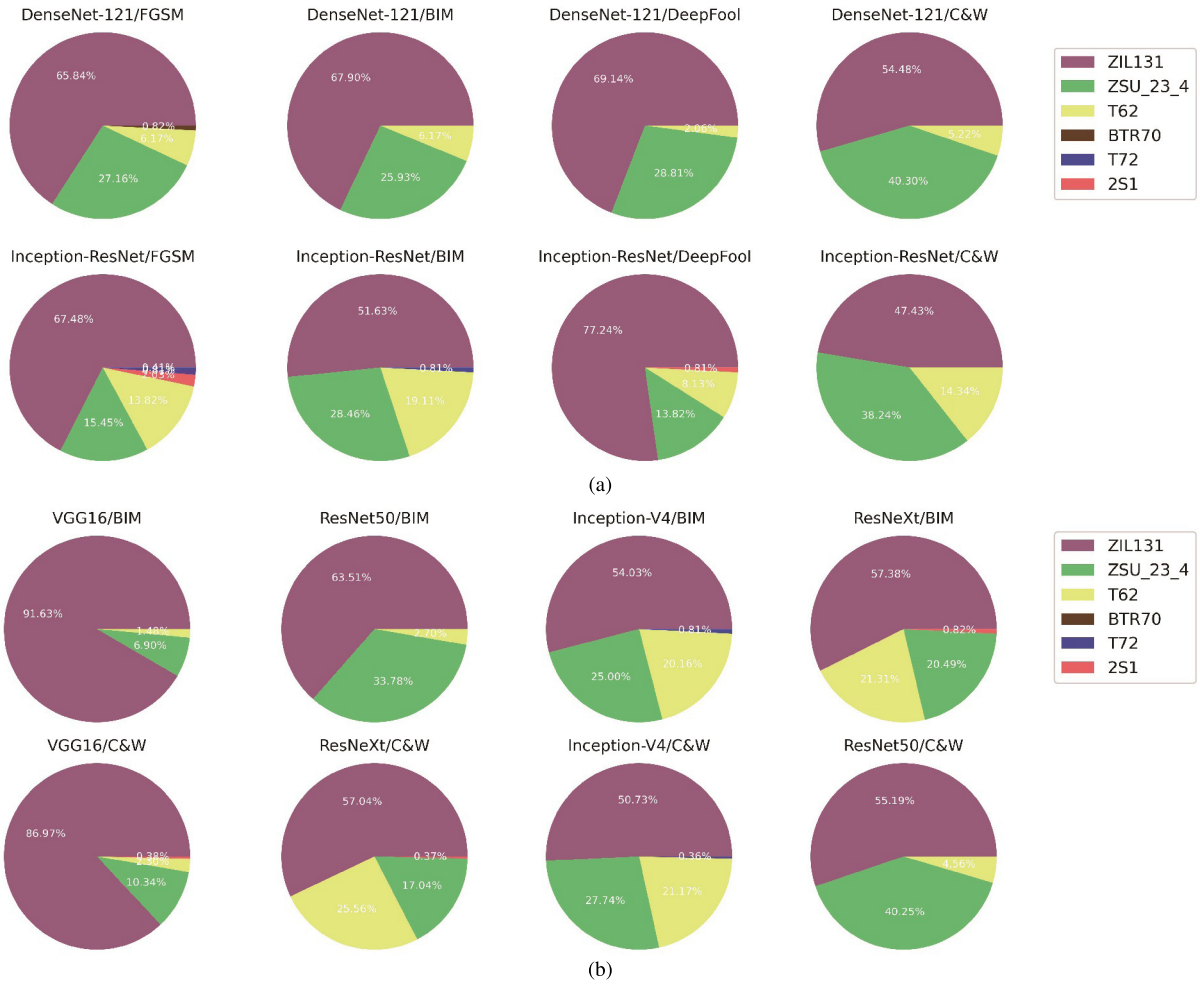


Fig. 9. Class distributions of adversarial examples in the class of D7 of the MSTAR data set. (a) Class distribution of the Inception-ResNet and DenseNet-12 model in different attacks. (b) Class distribution of the BIM and C&W attack algorithm across different models.

the classes of these adversarial examples are very different from the original classes of the objects. These cases have little impact on adversarial examples.

On the other hand, how to defend these adversarial examples in the field of remote sensing is also an important topic. In this regard, we can explore it from two perspectives. The first kind of defense algorithms could be designed according to our discoveries. For example, in optical RSIs, we can analyze and warn high-risk RSIs based on feature information or the distribution of adversarial examples, thereby designing defense algorithms in the field of remote sensing. The second is to apply defense algorithms in the field of computer version or machine learning (ML) in the field of remote sensing. These defense algorithms can be divided into four categories [15], [53], adversarial training [54], [55], gradient masking [56], [57], input transformation [58], and detection only [15]. These methods in the ML field can be applied to the field of remote sensing. However, their performance in the field of remote sensing may need further discussion.

IV. Conclusion

Adversarial examples are recognized as an urgent issue in the remote sensing community [22], [59], [60]. In this study, we conduct a comprehensively investigate adversarial example problem for RSI scene classification by applying several state-of-the-art DNNs for different classification scenarios. Through the empirical study, we have four conclusions.

1) Our result shows that all the DNN models can achieve an average fooling rate of 98% or above, which shows the universality and severity of adversarial example problems in RSI scene classification than we thought.
2) We find that adversarial examples have different impacts on different RSI data sets. For optical data sets, adversarial examples would divert little from the original images because they contain rich information in terms of spectral bands, the spatial dimension, and a high resolution. However, this rule doesn't apply to SAR data sets.
3) Multiple DNNs can be easily fooled with an average fooling rate of 76.01% for the SAR data sets. Adversarial examples generated under SAR data sets pose a greater threat to other classification models.
4) We also find that the classes of adversarial examples are more affected by the class distribution of data than the attack algorithms or the model architectures.

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These conclusions are useful for understanding adversarial examples in the field of remote sensing and the design of defense algorithms to solve the problem of adversarial examples.

There are some potential future works.

1) The adversarial perturbation covering the whole image cannot be applied in the real scene with various sizes of ground objects because we use smaller RSI patches as the input. Thus, we need to develop a new strategy to fool the classification model in a real scene.
2) Errors in the RSIs might come from the sensors, and multiview distortions of ground objects might also be problematic since the object might have adversarial perturbations when the same ground object is misclassified in several views. In the future, we would like to explore the relationship between the classification objects and their environment to help defend adversarial examples. Besides, we would also focus on the explanation of adversarial examples [45] from RSIs classification, detection, and segmentation tasks.

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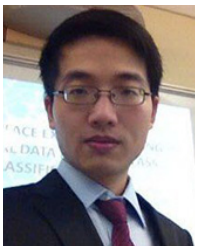
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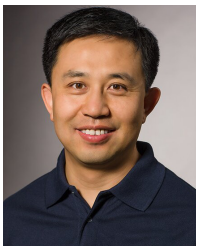
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